VBF-C1T1R1 calc - Virtual Battery Factory deliverable

CALC - VBF-C1T1R1-CALC-001

[Sheet: Calc]
Section I Quantity I Value I Unit I Source file
Input I capacity_ah I 5.03063266671673 I Ah I cell/r4_v3_margin_energy.json
Input I energy_wh I 17.89754980087133 I Wh I cell/r4_v3_margin_energy.json
Input I mass_kg I 0.0356025250776 I kg I cell/r4_v3_margin_energy.json
Input I area_m2 I 0.1027 I m2 I cell/r4_v3_margin_energy.json
Input I thickness_m I 0.0001828 I m I cell/r4_v3_margin_energy.json
Input I electrolyte_included I false I - I cell/r4_v3_margin_energy.json
Layer I positive_electrode layer_kg_m2 I 0.163993788 I kg/m2 I cell/r4_v3_margin_energy.json
Layer I negative_electrode layer_kg_m2 I 0.1058823 I kg/m2 I cell/r4_v3_margin_energy.json
Layer I positive_cc layer_kg_m2 I 0.0216 I kg/m2 I cell/r4_v3_margin_energy.json
Layer I negative_cc layer_kg_m2 I 0.05376 I kg/m2 I cell/r4_v3_margin_energy.json
Layer I separator layer_kg_m2 I 0.0014292 I kg/m2 I cell/r4_v3_margin_energy.json
Formula I ED = energy_wh / mass_kg I 17.8975498 / 0.0356025 = 502.7045 I Wh/kg I calc-energy contrac
Result I energy_density_wh_kg I 502.7045065444505 I Wh/kg I cell/r4_v3_margin_energy.json
Result I energy_density_wh_l I 953.3380882939265 I Wh/L I cell/r4_v3_margin_energy.json
Check I ED threshold I 392.61 I Wh/kg I entry-0 criteria stage2
Check I ED >= threshold I PASS (+110.09) I - I log-evaluate round 4